IB Mathematics Analysis & Approaches
Standard Level–Prediction Exams

Duration: 1:30 hrs **November 2026 - Paper 1** (No Calculator)

SECTION A

Question 1

[Maximum mark: 5]

Consider the following statement.

"The product of any two consecutive even integers is divisible by 4."

(a) Show that this statement is true for the integers 8 & 10. [2]

(b) Prove that the statement is true for any two consecutive even integers. [3]

(a) $$8 \times 10 = 80$$

which is divisible by 4

(b) Let $2n$ and $(2n+2)$ be two consecutive even integers

$$\therefore 2n \times (2n+2) = 4n^2 + 4n = 4\left(n^2 + n\right)$$

which is divisible by 4.

Question 2

Consider two events, A & B, such that

$$P(A) = y, \quad P(B) = 5y - 1, \quad P(A \cup B) = 0.65$$

(a) Find an expression for P($A \cap B$). [2]

(b) Given that events A and B are independent, find the value of $\cancel{y}$ y. [4]

(a)
$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$
$$= y + (5y - 1) - 0.65$$
$$P(A \cap B) = 6y - 1.65 \text{ ——— (i)}$$

(b) If A & B are independent, then

$$P(A \cap B) = P(A) \times P(B)$$

$\therefore$ (i) becomes

$$P(A) \times P(B) = 6y - 1.65$$
$$\therefore y(5y - 1) = 6y - 1.65$$
$$5y^2 - 7y + 1.65 = 0$$
$$y = \frac{7 \pm \sqrt{7^2 - 4(5)(1.65)}}{2(5)}$$
$$= \frac{7 \pm \sqrt{49 - 33}}{10}$$
$$= \frac{7 \pm 4}{10}$$
$$\therefore y = \frac{11}{10} \text{ or } y = \frac{3}{10}$$

But $P(A) = y \quad \therefore 0 \leq y \leq 1$ $\qquad \therefore y = \frac{3}{10}$

Question 3

[Maximum mark: 6]

(a) Express. $\log_2\left(\frac{4x^3}{y}\right)$ in terms of $\log_2 x$ and $\log_2 y$ [3]

(b) Hence, given that y = x² , solve the equation $\log_2\left(\frac{4x^3}{y}\right) = 5.$ [3]

(a) $$\log_2\left(\frac{4x^3}{y}\right) = \log_2\left(4x^3\right) - \log_2 y$$

$$= \log_2 4 + \log_2 x^3 - \log_2 y$$

$$= 2\log_2 2 + 3\log_2 x - \log_2 y$$

$$\log_2\left(\frac{4x^3}{y}\right) = 2 + 3\log_2 x - \log_2 y$$ $(\log_2 2 = 1)$

(b) $$\log_2\left(\frac{4x^3}{y}\right) = 5$$

$$\therefore 2 + 3\log_2 x - \log_2 y = 5$$ (by part(a))

$$y = x^2$$

$$\therefore 2 + 3\log_2 x - \log_2 x^2 = 5$$

$$\therefore 2 + 3\log_2 x - 2\log_2 x = 5$$

$$\therefore \log_2 x = 3$$

$$\therefore x = 2^3$$

$$\therefore x = 8$$

Question 4

[Maximum mark: 6]

Consider the function $f(x) = x^2 - 4x + 7$, where $x \in \mathbb{R}$.

(a) (i) Express $f(x)$ in the form $(x-h)^2 + k$.

(ii) Hence write down the coordinates of the vertex of the graph of f. [2]

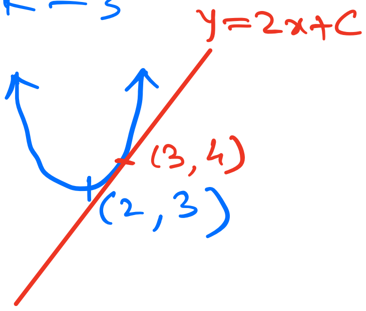
(b) The line $y = 2x + c$ is a tangent to the graph of f. Find the value of c. [3]

(c) Hence find the coordinates of the point where the line touches the graph of f. [1]

(a) (i) $f(x) = (x^2 - 4x + 4) - 4 + 7$

$\therefore f(x) = (x-2)^2 + 3$

$h=2, k=3$



(ii) vertex of graph of f is $(2,3)$

(b) $y = 2x + C$ is tangent to f

$$\therefore x^2 - 4x + 7 = 2x + C$$

$$\therefore x^2 - 6x + (7-C) = 0 \quad \text{—— (i)}$$

Now $\Delta = 0$

$$b^2 - 4ac = 0$$

$$\therefore (-6)^2 - 4(1)(7-c) = 0$$

$$-28 + 4C = -36$$

$$4C = -8$$

$$C = -2$$

(c) Hence find the coordinates of the point where the line touches the graph of f. [1]

Now equation (i) becomes

$$x^2 - 6x + (7 - (-2)) = 0$$

$$\therefore x^2 - 6x + 9 = 0$$

$$\therefore (x-3)^2 = 0$$

$$\therefore x = 3$$

$$\therefore y = 2x + (-2)$$

$$= 2(3) - 2$$

$$y = 4$$

Therefore, point of intersection is $(3, 4)$

Question 5

[Maximum mark: 7]

Consider the equation $\cos 2x + 3\sin x - 2 = 0$.

(a) Show that this equation can be written as $2\sin^2 x - 3\sin x + 1 = 0$. [2]

(b) Hence solve the equation $\cos 2x + 3\sin x - 2 = 0$ for $0 \le x \le 2\pi$, giving your answers in exact form. [4]

(c) Hence write down the number of solutions of the equation $\cos 2x + 3\sin x - 2 = 0$ for $-4\pi \le x \le 4\pi$. [1]

(a) $$\cos(2x) + 3\sin x - 2 = 0$$

$$\therefore (1 - 2\sin^2 x) + 3\sin x - 2 = 0$$

$$\therefore -2\sin^2 x + 3\sin x - 1 = 0$$

$$\therefore 2\sin^2 x - 3\sin x + 1 = 0$$

(Double angle formula)

$\cos 2x = \cos^2 x - \sin^2 x$
$= 2\cos^2 x - 1$
$= 1 - 2\sin^2 x$ ✓

(b) $\cos 2x + 3\sin x - 2 = 0$ for $0 \le x \le 2\pi$

$$\therefore 2\sin^2 x - 3\sin x + 1 = 0$$ (by part(a))

Let $y = \sin x$

$$\therefore 2y^2 - 3y + 1 = 0$$

$$\underline{2y^2 - 2y} - \underline{y + 1} = 0$$

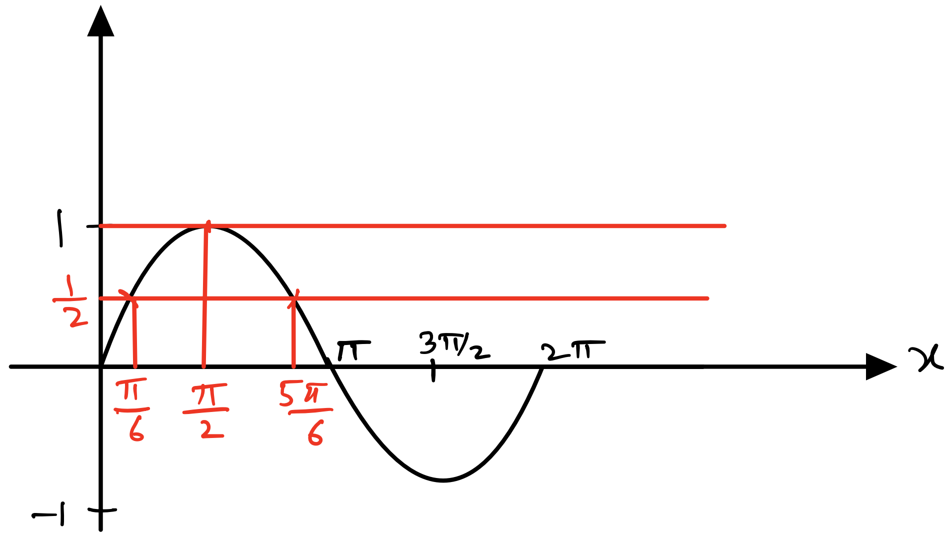
$$2y(y-1) - 1(y-1) = 0$$

$$\therefore (y-1)(2y-1) = 0$$

$$\therefore y = 1 \quad \text{or} \quad y = \frac{1}{2}$$

$$\therefore \sin x = 1 \quad \text{or} \quad \sin x = \frac{1}{2}$$

Plotting graph of $y = \sin x$ in $0 \le x \le 2\pi$



We get 3 solutions

For $\sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$

and for $\sin x = 1 \Rightarrow x = \frac{\pi}{2}$

(C) $\cos(2x) + 3\sin x - 2 = 0$ has period 2π in $0 \le x \le 2\pi$

Now in $-4\pi \le x \le 4\pi$. the function will have 4 more cycles.

$\therefore$ Number of solutions are

$$4 \times 3 = 12$$

In the interval $-4\pi \le x \le 4\pi$.

One Period
$0 \le x \le 2\pi$ (3 Solution)
In
$\underbrace{-4\pi \le x}_{2\ \text{Period}} \underbrace{\le 4\pi}_{2\ \text{Period}}$

Question 6

[Maximum mark: 7]

Consider the function $f(x) = \frac{\ln(\frac{x}{2})}{x}$, where $x \in \mathbb{R}^+$.

(a) Show that $f'(x) = \frac{1-\ln(\frac{x}{2})}{x^2}$. [2]

(b) Hence find the x-coordinate of the stationary point of the graph of f. [2]

(c) Determine the nature of the stationary point. [3]

(a) $$f(x) = \frac{\ln\left(\frac{x}{2}\right)}{x}$$

using quotient rule $y = \frac{u}{v} \Rightarrow y' = \frac{u'v - uv'}{v^2}$

$$f'(x) = \frac{\left(\frac{1}{x/2} \times \frac{1}{2}\right) x - \ln\left(\frac{x}{2}\right)(1)}{x^2}$$

$\left\{ \frac{d}{dx}\left(\ln\left(\frac{x}{2}\right)\right) = \frac{1}{(x/2)} \times \frac{1}{2} \right\}$

$$f'(x) = \frac{1-\ln\left(\frac{x}{2}\right)}{x^2}$$

(b) Taking $f'(x) = 0$, we get

$$1 - \ln\left(\frac{x}{2}\right) = 0$$

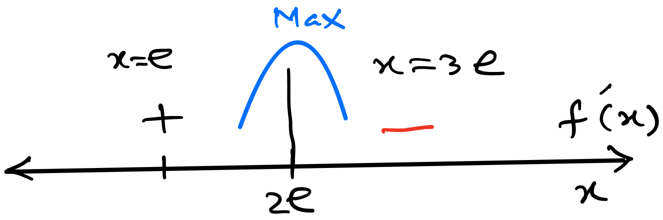
$$\therefore \ln\left(\frac{x}{2}\right) = 1$$

$$\ln\left(\frac{x}{2}\right) = \ln e$$

$$\therefore \frac{x}{2} = e \therefore x = 2e$$

(c) $\quad x = 2e \qquad f'(x) = \dfrac{1 - \ln\left(\frac{x}{2}\right)}{x^2}$

using sign diagram



$$f'(e) = \frac{1 - \ln\left(\frac{e}{2}\right)}{e^2}$$

$$= \frac{1 - \ln e + \ln 2}{e^2}$$

$$= \frac{\ln 2}{e^2} > 0$$

and $\quad f'(3e) = \dfrac{1 - \ln\left(\frac{3e}{2}\right)}{(3e)^2} = \dfrac{1 - (\ln(3e) - \ln 2)}{9e^2}$

$$= \frac{1 - (\ln 3 + \ln e - \ln 2)}{9e^2}$$

$$\therefore f'(3e) = \frac{\ln 2 - \ln 3}{9e^2} < 0$$

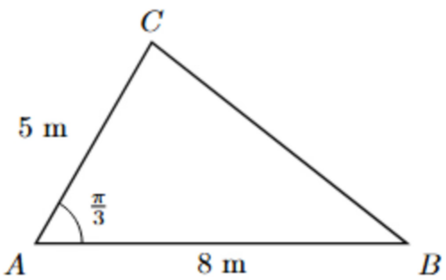
$\therefore x = 2e$ is a maximum point

SECTION B

Question 7

[Maximum mark: 16]

The following diagram shows a triangular patio ABC, which forms part of a garden. In the triangle, $AB = 8$ m, $AC = 5$ m and $B\hat{A}C = \frac{\pi}{3}$.



(a) Find the length of CB. [3]

(b) Find the area of the triangular patio ABC. [3]

(a) using cosine rule

$$CB^2 = 5^2 + 8^2 - 2(5)(8)\cos\frac{\pi}{3}$$

$$CB^2 = 49$$

$$\therefore CB = 7m$$

(b) $Area = \frac{1}{2}(5)(8)\sin\left(\frac{\pi}{3}\right) = 20\frac{\sqrt{3}}{2} = 10\sqrt{3}m^2$

The patio is paved with concrete to a uniform depth of 0.1 m.

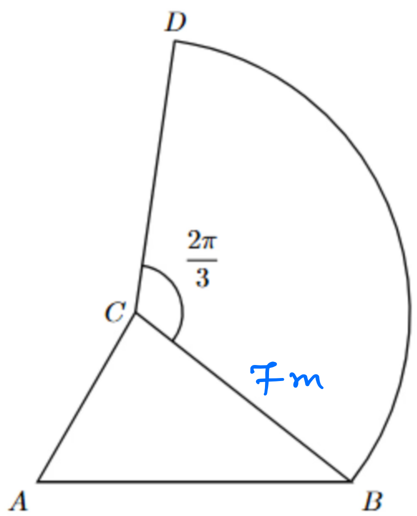
(c) Find the volume of concrete used for the patio. [2]

Volume = area × depth

$$= 10\sqrt{3} \times 0.1$$

$$= \sqrt{3}m^3$$

A sector-shaped lawn BCD is added to the garden. The lawn has centre C and radius CB, and the angle of the sector is $B\hat{C}D = \frac{2\pi}{3}$, as shown in the following diagram.



(d) Find the length of arc BD. [2]

By part (a) $CB = 7m$, $\theta = \frac{2\pi}{3}$

(Arc length $= r\theta$)

$\therefore$ arc length (BD) $= r\theta$

$$= 7 \times \frac{2\pi}{3}$$

$$= \frac{14\pi}{3}m$$

(e) Find the area of the lawn BCD. [2]

$$\text{area of } BCD = \frac{1}{2}r^2\theta$$

$$= \frac{1}{2}7^2\left(\frac{2\pi}{3}\right)$$

[4]

$$= \frac{49\pi}{3}m^2$$

A robot lawn mower is to be used to mow the lawn. The mower cutting blade has a width of 0.5 m, and can mow at a rate of $\frac{7\pi}{3}$ metres per minute.

(f) Determine the length of time the robot mower will take to mow the lawn. [4]

$$\text{Cutting rate} = 0.5 \times \frac{7\pi}{3} = \frac{7\pi}{6}\ m^2 \cdot \min^{-1}$$

$$\text{Time required} = \frac{\text{area}}{\text{cutting rate}}$$

$$= \frac{\left(\frac{49\pi}{3}\right)}{\left(\frac{7\pi}{6}\right)}$$

$$= 49\,\text{min}$$

Question 8

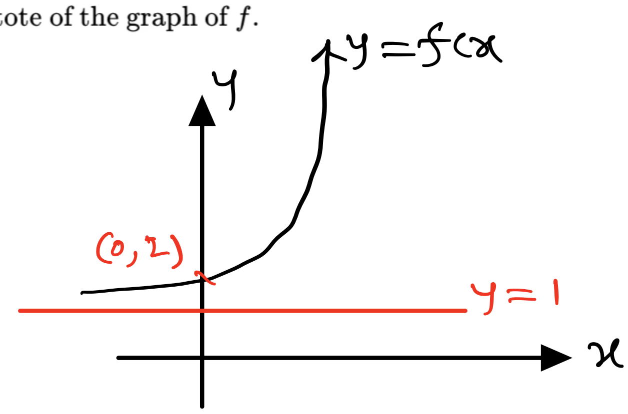
[Maximum mark: 14]

The function f is defined for $x \in \mathbb{R}$ by

$$f(x) = e^{2x} + 1$$

(a) (i) Write down the range of f.

(ii) Write down the equation of the horizontal asymptote of the graph of f. [2]



a(i) From graph

Range of f is

$$\{f(x) | 1 < y < \infty\}$$

(ii) Horizontal asymptote is

as $x \to -\infty, e^{2x} + 1 \to 1$

$\therefore y = 1$ is Horizontal asymptote

(b) Find $f^{-1}(x)$. [3]

$$f(x) = e^{2x} + 1$$

$$\therefore y = e^{2x} + 1$$

$$\therefore x = e^{2y} + 1$$

$$\therefore x - 1 = e^{2y} \Rightarrow 2y = \ln(x-1)$$

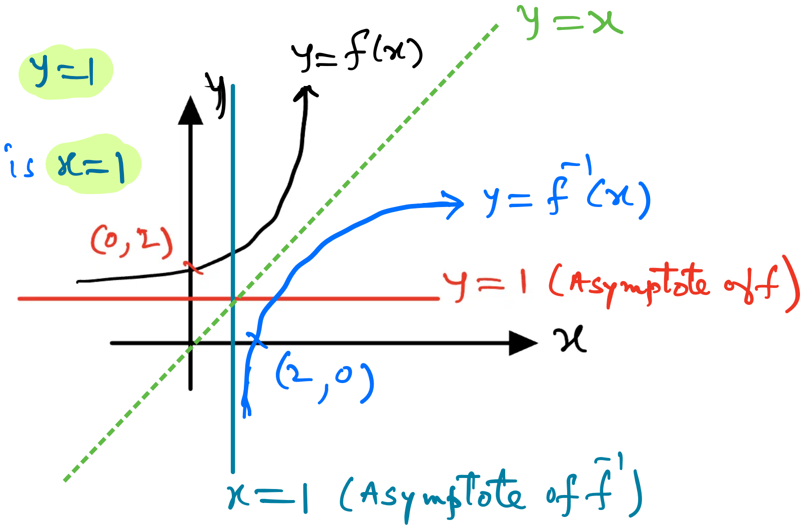
$$\therefore y = \frac{1}{2}\ln(x-1)$$

$$\therefore f^{-1}(x) = \frac{1}{2}\ln(x-1)$$

(c) On the same set of axes, sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$. Clearly indicate the asymptotes of each graph. [3]

Asymptote of $y=f(x)$ is $y=1$

Asymptote of $y=f^{-1}(x)$ is $x=1$



The function g is defined for $x \in \mathbb{R}$ by

$$g(x) = 3x - 2$$

(d) Find $(g \circ f)(x)$, giving your answer in its simplest form. [2]

(e) Hence solve the equation $(g \circ f)(x) = 7$. Give your answer in the form $a \ln b$, where $a, b \in \mathbb{Q}^+$. [4]

(d) $$(g \circ f)(x) = g(f(x)) = g\left(e^{2x} + 1\right) = 3\left(e^{2x} + 1\right) - 2$$

$$= 3e^{2x} + 1$$

(e) $$(g \circ f)(x) = 7$$

$$\therefore 3e^{2x} + 1 = 7$$

$$3e^{2x} = 6$$

$$e^{2x} = 2$$

$$2x = \ln 2$$

$$x = \frac{1}{2}\ln 2$$

Question 9

[Maximum mark: 13]

A curve C has gradient given by $\frac{dy}{dx} = 3x^2 + 2x - 1$, and C passes through the origin.

(a) Find the equation of C. [3]

Integrating both sides with respecting to 'x'

$$\int \frac{dy}{dx} dx = \int \left(3x^2 + 2x - 1\right) dx$$

$$\therefore y = \frac{3x^3}{3} + \frac{2x^2}{2} - x + k \quad \Rightarrow y = x^3 + x^2 - x + k, k \in R$$

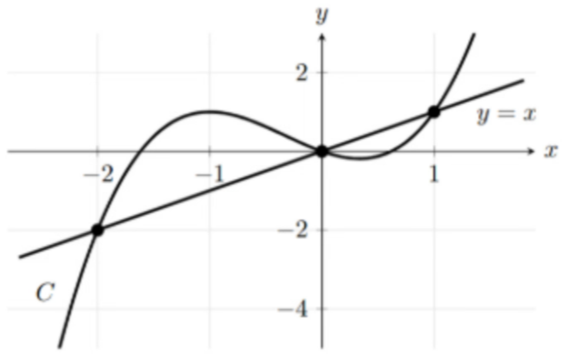
It passes through $(0,0)$

$$\therefore 0 = (0)^3 + (0)^2 - (0) + k \quad \Rightarrow k = 0$$

$\therefore$ equation of curve C is

$$y = x^3 + x^2 - x$$

The following diagram shows the curve C and the line $y = x$, which intersect at three points.



(b) Find the coordinates of the three points of intersection of C and the line $y = x$. [4]

$$x^3 + x^2 - x = x$$

$$\therefore x^3 + x^2 - 2x = 0 \quad \Rightarrow x\left(x^2 + x - 2\right) = 0$$

$$\therefore x\,(x+2)\,(x-1) = 0$$

$$\therefore x = -2, x = 0, x = 1$$

$$For x = -2, y = -2 \Rightarrow (-2, -2)$$ $\left(y = \frac{1}{x} \right)$

$$For x = 0, y = 0 \Rightarrow (0, 0)$$

$$For x = 1, y = 1 \Rightarrow (1, 1)$$

$\therefore$ $(-2, -2)$, $(0, 0)$ and $(1, 1)$ are points of intersection

A new curve D is formed by translating the curve C two units in the direction of the positive x-axis, and two units in the direction of the positive y-axis.

(c) Find the coordinates of the three points of intersection of D and the line $y = x$. [2]

(d) Find the total area of the region enclosed by D and the line $y = x$. [4]

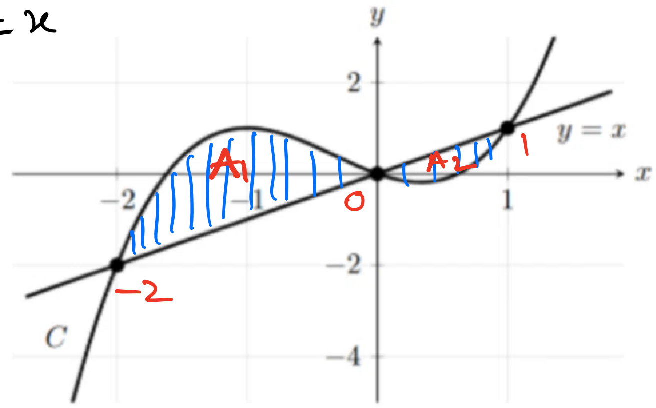
(c) $(-2, -2) \Rightarrow (-2 + 2, -2 + 2) \Rightarrow (0, 0)$

$$(0, 0) \Rightarrow (0 + 2, 0 + 2) \Rightarrow (2, 2)$$

$$(1, 1) \Rightarrow (1 + 2, 1 + 2) \Rightarrow (3, 3)$$

$\therefore (0, 0), (2, 2), (3, 3)$ are points of intersection of D and line $y = x$.

(d) The area between C and $y=x$ is congruent to area between D & $y=x$



Let A_1 area between D & $y=x$ from $x=-2$ to $x=0$

$$\therefore A_1 = \int_{-2}^{0} (upper - lower)\, dx = \int_{-2}^{0} (x^3 + x^2 - x) - (x)\, dx$$

$$= \int_{-2}^{0} x^3 + x^2 - 2x dx$$

$$= \left[\frac{x^4}{4} + \frac{x^3}{3} - x^2\right]_{-2}^{0}$$

$$\therefore A_1 = 0 - \left(\frac{(-2)^4}{4} + \frac{(-2)^3}{3} - (-2)^2\right) = \left(-4 + \frac{8}{3} + 4\right)$$

Let A_2 be area between D & $y=x$ from $x=0$ & $x=1$

$$\therefore A_2 = \int_0^1 x - \left(x^3 + x^2 - x\right) dx$$

$$= \int_0^1 \left(2x - x^3 - x^2\right) dx$$

$$= \left[x^2 - \frac{x^4}{4} - \frac{x^3}{3}\right]_0^1$$

$$\therefore A_2 = \left(1 - \frac{1}{4} - \frac{1}{3}\right)$$

∴ Total area A is

$$A = A_1 + A_2$$

$$= \left(\cancel{-4} + \frac{8}{3} + \cancel{4}\right) + \left(1 - \frac{1}{4} - \frac{1}{3}\right)$$

$$\therefore A = \frac{37}{12} \text{ units}^2$$

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